Generative Adversarial Networks for Data Augmentation and Domain Adaptation

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Abstract

Class imbalance and the scarcity of labeled data limit the effectiveness of Deep Learning models in medical diagnostics. This work proposes a framework to improve pneumonia classification in pediatric X-rays [9] by integrating generative Data Augmentation and adversarial Domain Adaptation techniques. In this work, two conditional generative models are compared: a cDC-WGAN-GP [1, 5] and a cDC-SNGAN [12], both equipped with a PatchGAN discriminator [8] for local details and a balanced sampling strategy inspired by BAGAN [10]. The metric quality of the samples is measured via FID [7] and KID [2]. The results show that adding synthetic images for the minority class (NORMAL) trains a more robust ResNet18 [6], optimizing the Macro F1-score. To visualize the latent space and quantify the domain shift, we use dimensionality reduction techniques (PCA, t-SNE, UMAP). Finally, we integrate a DANN architecture [4] to align the real and synthetic latent distributions: in a controlled comparison the adversarial alignment provides a modest but consistent gain on the minority class, while the latent-space analysis shows that the real-synthetic separation is only attenuated and not fully removed. As a further step towards clinical generalization, we outline a cross-hospital domain adaptation scenario, in which the classifier is trained and evaluated on radiographs acquired from different sources.

1. Introduction

The automated interpretation of chest X-rays represents a pillar of computer-aided diagnostics. However, the development of robust models is hindered by class imbalance in real-world datasets. The "Chest X-Ray Pneumonia" pediatric dataset [9], for example, presents an unbalanced ratio of approximately 1:3 between healthy and pathological cases. In this scenario, we evaluate a classifier based on ResNet18 [6], with fine-tuning limited to the last convolutional blocks (`layer3`, `layer4`) and the linear head. This setup tends to maximize global accuracy at the expense of recall on the healthy class, resulting in a sub-optimal Macro F1-score.

To mitigate this imbalance, we propose a generative Data Augmentation strategy comparing two conditional architectures: a cDC-WGAN-GP [1, 5] and a cDC-SNGAN [12]. We evaluate these models both in their standard formulations and when enhanced with a PatchGAN discriminator [8] and a BAGAN-inspired sampling strategy [10]. The generated synthetic samples are then evaluated quantitatively via FID [7] and KID [2] to certify their alignment with real images.

However, the insertion of synthetic data introduces a *domain gap* that can push the classifier to learn shortcuts related to generation artifacts. To map and visualize this phenomenon, we project the ResNet18 feature space using dimensionality reduction techniques (PCA, t-SNE, and UMAP). To align the latent distributions, we integrate a Domain Adaptation pipeline based on the Domain-Adversarial Neural Network (DANN) architecture [4]. Through a Gradient Reversal Layer, this network forces the feature extractor to generate domain-invariant representations, bridging the gap between real and generated images. The resulting framework combines generative augmentation, quantitative validation, and adversarial adaptation. Our analysis shows that generative augmentation effectively improves minority-class performance. Furthermore, applying a mild adversarial adaptation attenuates the residual real-synthetic gap. Although this gap is not fully closed, the adaptation yields a small additional gain on the minority class, which we analyze through latent-space visualizations. As a further direction towards clinical generalization, we also outline a cross-hospital domain adaptation scenario, in which radiographs from different sources are treated as distinct domains.

2. Related Work

Recent literature on medical imaging integrates convolutional models and generative networks to address data scarcity.

2.1. Residual Classification in Radiology

Modern image recognition is driven by the Residual Networks (ResNet) introduced by He et al. [6]. Their shortcut connections allow for the effective training of very deep architectures. In the radiographic field, Kermany et al. [9] introduced the dataset under examination, demonstrating the effectiveness of transfer learning to assist clinicians in pediatric pneumonia classification. However, model performance collapses in the presence of class imbalances, pushing research towards robust data balancing and augmentation techniques.

2.2. Data Augmentation via Generative Models

Generative Adversarial Networks (GANs) have revolutionized image generation. DCGANs [13] and their conditional extension cGAN [11] have made targeted generation guided by class labels possible. To counteract the instability of adversarial training, the Wasserstein GAN (WGAN) [1] replaced the Jensen-Shannon divergence with the Earth-Mover distance. This formulation was subsequently optimized via Gradient Penalty (WGAN-GP) [5]. Alternatively, Spectral Normalization (SNGAN) [12] directly constrains the Lipschitz constant of the discriminator. It achieves this by normalizing weights with respect to their maximum singular value, thereby reducing computational cost through Hinge loss. Furthermore, to preserve high-frequency textures crucial in X-rays (such as ribs and parenchyma), Isola et al. [8] introduced the PatchGAN discriminator, which evaluates local patches instead of the entire image. Finally, to handle unbalanced datasets, the BAGAN framework [10] demonstrated how latent regularization and batch balancing can effectively prevent mode collapse towards the majority classes.

2.3. Adversarial Domain Adaptation

The statistical discrepancy (*domain gap*) between real and synthetic images introduces structural biases. These biases can induce the classifier to learn generative artifacts rather than robust clinical patterns. Domain Adaptation techniques aim to mitigate this gap. The reference paradigm is represented by the Domain-Adversarial Neural Networks (DANN) of Ganin et al. [4]. Through a Gradient Reversal Layer (GRL), DANNs align real and synthetic latent distributions during adversarial training. This forces the feature extractor to produce representations that are invariant with respect to the source domain. This mechanism constitutes the foundational principle upon which our pipeline rests.

3. Methodology

The core of this project is a sequential pipeline designed to validate a two-fold hypothesis. First, targeted synthetic generation can mitigate the structural limitations of an unbalanced dataset. Second, the resulting domain gap can be effectively bridged through adversarial adaptation.

3.1. Baseline Evaluation and Imbalance Analysis

The starting point of this work consists of establishing a reference benchmark by training a ResNet18 classifier on the original dataset. This model is trained using a partial fine-tuning strategy. Specifically, all backbone parameters are frozen, with the exception of `layer3` and `layer4`. The final fully-connected layer is then replaced with a new binary classifier. This choice preserves the low-level generic features previously learned on ImageNet. Consequently, it concentrates the adaptation capacity on the high-level representations most relevant to the radiographic domain. Since global accuracy is misleading on imbalanced data, we adopt the Macro F1-Score as our primary metric. As the unweighted mean of the per-class F1 scores, it cannot be inflated by simply favouring the majority (PNEUMONIA) class. Instead, it requires robust precision and recall on the minority NORMAL class as well. To match the input requirements of the ImageNet pre-trained backbone, all single-channel grayscale X-rays are converted to RGB. This is achieved by replicating the single channel three times. Finally, all images (both real and generated) are resized to 224×224 pixels.

3.2. Implementation of the cDC-WGAN-GP

To bridge the class imbalance, we first implement a cDC-WGAN-GP. Its conditional design allows for class-targeted generation. We adopt the Wasserstein loss with Gradient Penalty (GP) to enforce the 1-Lipschitz constraint $\|\nabla_{\hat{x}} D(\hat{x})\|_p \leq 1$ and avoid mode collapse. To keep the gradient penalty well-behaved, we replace Batch Normalization in the Critic with Instance Normalization. Additionally, we introduce an Epsilon Drift Penalty to prevent logit drift during late training stages.

3.3. Implementation of the cDC-SNGAN

Alternatively, we implement a cDC-SNGAN [12], which achieves training stabilization structurally. Each convolutional layer of the Critic is constrained via Spectral Normalization (as detailed in Section 2). This technique imposes the Lipschitz constraint directly without requiring additional penalty terms. The objective function adopted for this model is the Hinge Loss:

$$\begin{aligned} L_D &= -\mathbb{E}_{x\sim p_{\text{data}}}[\min(0, -1 + D(x))] \\ &\quad - \mathbb{E}_{z\sim p_z}[\min(0, -1 - D(G(z)))] \\ L_G &= -\mathbb{E}_{z\sim p_z}[D(G(z))] \end{aligned} \tag{1}$$

This formulation significantly reduces computational costs. It allows the Critic to be updated only once per step

($n_{\text{critic}} = 1$), compared to the five updates ($n_{\text{critic}} = 5$) required by the cDC-WGAN-GP. Both generative models are evaluated at the standard resolution of 128×128 pixels and at a higher resolution of 256×256 pixels to analyze the impact of spatial detail.

3.4. High-Frequency Texture Preservation via PatchGAN

Standard GAN discriminators typically output a single global scalar, which can sometimes lead to blurry generations. To preserve the high-frequency biological textures crucial in medical imaging (such as ribs and lung parenchyma), both our generative models integrate a PatchGAN discriminator [8]. Instead of evaluating the image globally, this network is configured with a 16×16 output grid. Its 256 local judgments force the generator to reproduce coherent and realistic local details, significantly improving the clinical fidelity of the synthetic X-rays.

3.5. Generative Balancing Strategy

Inspired by the BAGAN framework [10], we implement a specific balancing strategy during the training phase of both GANs. The Generator is forced to produce balanced batches featuring a 50/50 distribution between the NORMAL and PNEUMONIA classes:

$$\mathbf{y}_{\text{gen}} = \text{shuffle}\left(\underbrace{[0,\ldots,0]}_{\lfloor B/2 \rfloor} \| \underbrace{[1,\ldots,1]}_{B-\lfloor B/2 \rfloor}\right) \tag{2}$$

Here, B denotes the mini-batch size. The Critic is explicitly trained on the unbalanced real distribution without oversampling, allowing it to learn the true prior of the classes. Conversely, the Generator is forced to produce balanced batches to prevent mode collapse induced by the majority class.

3.6. Evaluation of Generative Quality

To quantify the statistical fidelity of the synthetic samples, we employ two standard metrics. The first is the Fréchet Inception Distance (FID) [7]. FID measures the Wasserstein distance between the distributions of real and synthetic features extracted from an Inception-v3 network (Eq. 3):

$$\text{FID} = |\mu_r - \mu_g|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right) \tag{3}$$

The second metric is the Kernel Inception Distance (KID) [2]. KID provides an unbiased estimator based on the Maximum Mean Discrepancy (MMD), making it particularly suitable for limited-size datasets.

3.7. Analysis of the Feature Space via PCA, t-SNE and UMAP

We aim to visualize the domain gap and understand how the ResNet18 organizes its internal representations. To achieve this, we project the 512-dimensional feature space into two dimensions using three standard techniques. These include **PCA** (linear, capturing global variance), **t-SNE** (non-linear, emphasizing local structure), and **UMAP** (non-linear, balancing local and global topology). This comparative visualization makes the structural domain shift explicit. Consequently, it strongly motivates the subsequent Domain Adaptation step.

3.8. Real-Synthetic Domain Adaptation

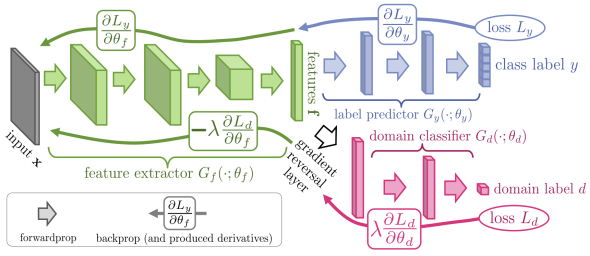


Figure 1. Architecture of the DANN model for clinical classification and adversarial domain adaptation.

To align the real and synthetic distributions, we must prevent the classifier from learning the *shortcut features* described in Section 2. Therefore, we integrate the DANN adversarial architecture [4] into our pipeline. This architecture consists of four fundamental elements, as illustrated in Figure 1:

- **Feature Extractor** (G_f): based on ResNet-18, maps input samples into the latent feature space.
- **Label Predictor** (G_y): performs clinical classification (healthy vs pneumonia) on the extracted features.
- **Domain Discriminator** (G_d): analyzes the latent space to distinguish the domain of origin (real vs synthetic) of the sample.
- **Gradient Reversal Layer (GRL)**: interposed between G_f and G_d, behaves as identity in the forward propagation and reverses the gradient by scaling it by $-\lambda$ in the backpropagation.

In this setup, we designate the real data as the *Source* domain and the generated data as the *Target* domain. The overall loss function of the adaptation system is thus defined as:

$$E(\theta_f, \theta_y, \theta_d) = \sum_{i \in \text{Real}} L_y(G_y(G_f(x_i)), y_i) + \alpha_{\text{synth}} \sum_{j \in \text{Synth}} L_y(G_y(G_f(x_j)), y_j) - \lambda \sum_{i \in \text{Real, Synth}} L_d(G_d(G_f(x_i)), d_i) \quad (4)$$

Here, L_y represents the clinical classification loss, while L_d denotes the domain discriminator loss. The parameter α_{synth} controls the weight associated with the supervision on synthetic data. Optimizing this minmax objective forces the Feature Extractor to generate representations that are invariant with respect to the input domain. Ultimately, this ensures a robust and unbiased clinical classification of real patient samples.

4. Experimental Setup

For reproducibility, we report below the main technical parameters.

4.1. Dataset and Validation Set Rebalancing

We use the "Chest X-Ray Pneumonia" dataset [9] of pediatric radiographs (NORMAL and PNEUMONIA). Its official validation split has only 16 images (8 per class), too few for stable metrics, so we rebuilt training and validation with a stratified re-sampling, leaving the test set untouched (Table 1).

Class	**Train**	**Validation**	**Test**	**Total**
Original Dataset				
NORMAL	1341	8	234	1583
PNEUMONIA	3875	8	390	4273
Total	5216	16	624	5856
Repartitioned Dataset				
NORMAL	1144	205	234	1583
PNEUMONIA	3503	380	390	4273
Total	4647	585	624	5856
Percentage	79.35%	9.99%	10.66%	100%

Table 1. Class distribution for the different dataset splits before and after stratified re-sampling.

4.2. Classifier Training

All ResNet18 classifiers, whether trained exclusively on real images or on the augmented dataset, follow the partial fine-tuning strategy described in Section 3. Training is conducted for 10 epochs with a batch size of 32 and a learning rate of 10^{-4} via the Adam optimizer.

4.3. GAN Optimization Dynamics

Both generative architectures (WGAN-GP and SNGAN) are trained on images standardized at a resolution of 128×128 pixels for 300 epochs, with a batch size of 64, a latent space dimension $n_z = 100$, and a base channel width $d = 128$. For SNGAN, an additional experiment is conducted at a resolution of 256×256 pixels.

We use Adam with an initial learning rate of 10^{-4}, reduced to 5×10^{-5} only for the SNGAN at 256×256 for stability, and a LinearLR scheduler that decays it towards zero. The two models differ as follows:

- **WGAN-GP**: The Critic is updated with frequency $n_{\text{critic}} = 5$ relative to the Generator. Adam parameters are set to $\beta_1 = 0.0$ and $\beta_2 = 0.9$ to allow for rapid adaptations as the Critic varies. Additionally, a weight decay of 10^{-3} and an Epsilon Drift Penalty coefficient of 10^{-3} are applied to the Critic.
- **SNGAN**: The discriminator is updated with frequency $n_{\text{critic}} = 1$, benefiting from Spectral Normalization regularization and Hinge Loss stability.

4.4. DANN: Training and Optimization Setup

Training of the DANN architecture for real-synthetic alignment is conducted for 50 epochs adopting balanced inter-domain sampling, in which each mini-batch of 64 samples is composed of 32 real (*Source*) images and 32 synthetic (*Target*) ones. To isolate the contribution of the adversarial term, we run two configurations that share the same initialization, data (75% synthetic quota) and schedule, differing *only* in the Gradient Reversal Layer strength: a control with $\lambda = 0$ (adversarial alignment disabled) and the DANN run with λ annealed from 0 to 1. The contrast between the two runs is reported in Section 5.6.

4.4.1. Architectural Details

The Feature Extractor (G_f) is based on a ResNet-18 pretrained on ImageNet, of which only the `layer3` and `layer4` blocks are unlocked; the output, following an Average Pooling layer, returns 512-dimensional latent vectors. The Label Predictor (G_y) is composed of a single-layer linear fully-connected network that maps the latent feature space to clinical classes ($512 \rightarrow 2$). The Domain Discriminator (G_d) is structured as a three-level MLP ($512 \rightarrow 256 \rightarrow 128 \rightarrow 1$), in which hidden layers adopt the ReLU activation function and 30% Dropout (0.3) to prevent overfitting.

4.4.2. Loss Functions and GRL

The clinical classification task loss is calculated via CrossEntropyLoss by combining two contributions: a component on real samples with a weight equal to 1.0, and a component on synthetic samples weighted by $\alpha_{\text{synth}} = 1.0$; real and synthetic labels therefore contribute equally to the

supervised term (Supervised Domain Adaptation). The domain loss makes use of BCEWithLogitsLoss applied to both domains (label 0 for real, 1 for synthetic). To guide adversarial backpropagation, the λ coefficient of the Gradient Reversal Layer is dynamically modulated during training according to the formulation proposed by Ganin et al. [4]:

$$\lambda = \frac{2}{1 + e^{-10p}} - 1 \tag{5}$$

where $p \in [0, 1]$ represents training progress (fraction of the total number of epochs). This setting causes λ to grow gradually from 0 to 1, allowing the Feature Extractor to first consolidate discriminative features for clinical classification, and then force domain alignment in the advanced phases.

4.4.3. Optimization and Model Selection

We use Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$) with a differential learning rate to protect the pre-trained filters: 10^{-4} for the Feature Extractor and 10^{-3} for the Label Predictor and the Domain Discriminator. The best checkpoint is selected on the Macro F1 of the real-only validation set, not on the total loss, so that model selection reflects clinical utility rather than synthesis noise.

5. Results and Discussion

This section reports our findings on three aspects: the quality of the generated images, the effect of generative balancing against standard baselines, and the impact of adversarial Domain Adaptation (DANN) on the residual real-synthetic gap.

5.1. Analysis of Generative Metrics and Visual Quality

We assess the quality of the generated samples with the Fréchet Inception Distance (FID) and the Kernel Inception Distance (KID). Table 2 reports both metrics at convergence, with the corresponding epoch.

Model	**WGAN-GP** (128×128)		**SNGAN** (128×128)		**SNGAN** (256×256)	
Variant	Standard	Complete	Standard	Complete	Standard	Complete
Epoch	260	300	170	210	110	170
FID	172.87	264.63	127.85	98.67	187.64	184.11
KID	0.2356	0.3590	0.1720	0.1300	0.2780	0.2640

Table 2. Generative metrics (FID, KID) and number of epochs for the GAN models under examination.

In the experimental study, each model is evaluated via an ablation study that provides for two configurations:

- **Standard**: the basic model configuration, devoid of local constraints and balancing strategies.
- **Complete**: the proposed variant that integrates both the PatchGAN discriminator and the BAGAN-like balanced sampling.

The effect of the Complete modules is architecture-dependent. On the WGAN-GP they worsen both metrics and lengthen training: the PatchGAN constraint and the 50/50 sampling create a trade-off between global alignment and minority-class stability that this model does not reconcile. On the SNGAN, instead, the Complete configuration acts as a regularizer and improves both metrics at each resolution, with the best result at 128×128 (FID 98.67, KID 0.1300).

Qualitatively (Figure 2), the PatchGAN discriminator pushes the model towards locally coherent textures, such as the lung parenchyma and the rib profile, reducing the blurring and the artifacts of the Standard models, while the BAGAN-like balancing preserves diversity in the minority class.

5.2. Impact of Classical Geometric Augmentation

Before evaluating the generative models, we tested classical geometric augmentation as a baseline. Following clinical recommendations for chest X-rays [3], we discarded transformations that produce anatomically implausible images (large rotations, reflections, shearing, strong scaling) and kept only mild rotations ($\pm 5°$), translations ($\pm 5\%$) and scaling ($\pm 5\%$), which reproduce differences in patient positioning and detector alignment. We trained the ResNet-18 while varying the fraction of augmented images from 25% to 100% (Table 3).

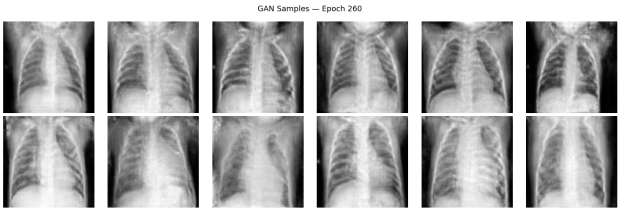
Table 3. Ablation study on F1-score as the proportion (%) of classical geometric augmentation varies.

%	**Class**	**Precision**	**Recall**	**F1-Score**
25%	N	0.99	0.47	0.63
	P	0.76	1.00	0.86
50%	N	0.99	0.56	0.72
	P	0.79	1.00	0.88
75%	N	0.98	0.51	0.67
	P	0.77	0.99	0.87
100%	N	0.98	0.54	0.69
	P	0.78	0.99	0.87

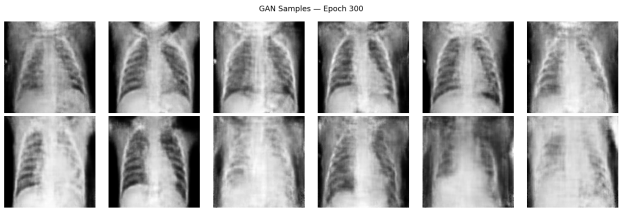
Classical transformations give only a mild regularizing effect, best at the 50% rate (Normal F1 0.72), and degrade at higher quotas because of overfitting on a limited set of patterns. They do not add real morphological diversity, which motivates the generative approach.

5.3. Ablation Study and Impact on Classification

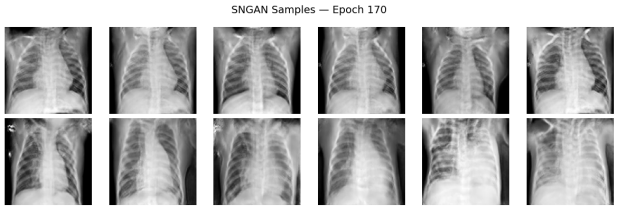
We then measured the effect of synthetic images as augmentation, training the ResNet-18 on the real training set enriched with a variable quota of synthetic NORMAL samples (25% to 100% of the class gap) and testing on the real



(a) WGAN-GP (128 × 128) Standard (epoch 260).



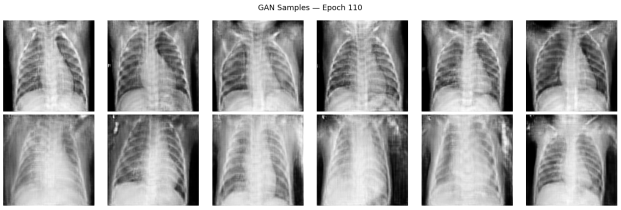
(b) WGAN-GP (128 × 128) Complete (epoch 300).



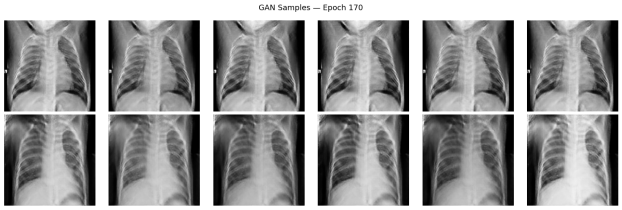
(c) SNGAN (128 × 128) Standard (epoch 170).



(d) SNGAN (128 × 128) Complete (epoch 210).



(e) SNGAN (256 × 256) Standard (epoch 110).



(f) SNGAN (256 × 256) Complete (epoch 170).

Figure 2. Qualitative comparison of NORMAL samples generated at 128 × 128 and 256 × 256 pixel resolutions between the Standard (left) and Complete (right) variants.

test set. N and P denote the Normal (minority) and Pneumonia (majority) classes (Table 4).

Table 4. Ablation study on F1-score as the quota (%) of synthetic data varies.

%	Class	**WGAN-GP** (128 × 128)		**SNGAN** (128 × 128)		**SNGAN** (256 × 256)	
		Standard	Complete	Standard	Complete	Standard	Complete
25%	N	0.70	0.70	0.70	0.74	0.72	0.72
	P	0.88	0.88	0.88	0.89	0.88	0.88
50%	N	0.67	0.68	0.66	0.74	0.72	0.71
	P	0.87	0.87	0.87	0.89	0.88	0.88
75%	N	0.56	0.70	0.65	**0.75**	0.64	0.63
	P	0.84	0.88	0.87	**0.89**	0.86	0.86
100%	N	0.55	0.70	0.68	0.64	0.63	0.66
	P	0.84	0.88	0.87	0.86	0.86	0.87

Generative augmentation outperforms the other balancing strategies (Table 5): class weighting and random oversampling give only modest gains (Normal F1 ≈ 0.62) and classical augmentation is the best non-generative baseline (Normal F1 0.72), while the Complete SNGAN (128×128) reaches the best Normal F1 of 0.75.

Two points stand out. First, a partial 75% quota is better than fully closing the gap (100%): the excess of synthetic samples biases the classifier and lowers the Normal F1. Second, for the SNGAN (128 × 128) the Complete variant is at once the most faithful (best FID and KID) and the most useful downstream, raising the Normal F1 from 0.65 to 0.75 at the 75% quota.

Table 5. Comparison of F1-score on the test set between baselines, classic methods, and the best GAN configuration (75%).

Methodology / Configuration	**Normal (N)**	**Pneumonia (P)**
Baseline (Real Only)	0.578	0.847
ClassWeight	0.626	0.857
Oversampling (UpSampling)	0.6257	0.8587
Classical Augmentation (50%)	0.720	0.880
Best GAN (SNGAN 128 × 128 **Complete, 75%)**	**0.750**	**0.890**

Across decision thresholds, the baseline and the best GAN configuration have almost identical macro-AUC (0.9528 vs 0.9538; Figure 3, Table 6) despite the clear F1 gain. AUC is threshold-independent and measures ranking, which the baseline already has; what it lacks is a well-calibrated boundary at the fixed 0.5 threshold. Augmentation rebalances the training distribution and moves the operating point towards the minority class, improving Normal recall and F1 without changing the ranking. The result is a flat AUC but a higher F1-score.

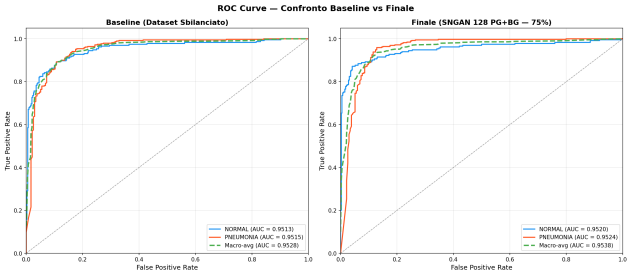


Figure 3. Comparison of ROC curves evaluating the impact of generative data augmentation on the classification task.

Table 6. Comparison of ROC-AUC scores between Baseline and Best GAN configuration.

Class	AUC Baseline	AUC Best GAN	Δ AUC
Normal (N)	0.9513	0.9520	+0.0007
Pneumonia (P)	0.9515	0.9524	+0.0009
Macro Average	0.9528	0.9538	+0.0010

5.4. Feature Visualization and Domain Distinguishability

To see how the ResNet-18 organizes real and synthetic samples, we extracted the 512-dimensional features from the penultimate layer and projected them into two dimensions with PCA, t-SNE and UMAP (Figure 4).

All three projections tell the same story: the network separates the three groups (Real Normal, Real Pneumonia and Fake Normal) very clearly, and the synthetic samples form an isolated cluster that does not overlap with the real NORMAL images. This is a diagnostic, not a pathological, finding. The GAN has not removed the domain shift, yet the synthetic cluster still carries enough clinical structure to improve the Normal F1 on the real test set, which is why augmentation helps despite the visible gap. The persistent separation motivates the Grad-CAM check (Section 5.5) and the adversarial alignment (Section 5.6).

5.5. Interpretability via Grad-CAM

To verify that the classifier relies on anatomically meaningful regions rather than on generation artifacts or acquisition markers (a well-documented shortcut in chest X-ray models), we computed Grad-CAM saliency maps on the last convolutional block of the ResNet-18, comparing the baseline (real-only) and the augmented (75%) classifiers on the same samples (Figure 5).

The maps support two observations. First, on real images the augmented classifier concentrates on the lung fields, while the baseline more often activates on peripheral regions such as the image borders and the laterality marker; this suggests that generative augmentation reduces the reliance on non-anatomical shortcuts. Second, and most important for our purpose, the synthetic NORMAL images do not show stronger activations at the borders than the real NORMAL images: their saliency also falls on plausible thoracic anatomy, and they are classified with high confidence. Since the classifier does not attend to the peripheral regions where GAN artifacts and acquisition markers would appear, we find no evidence that it relies on generation shortcuts to recognize the synthetic samples. These findings provide important context to the feature-space analysis. While the network is clearly able to separate the real and synthetic domains (likely due to low-level generation artifacts or textures), the Grad-CAM maps confirm that this domain shift does not distract the classifier from the anatomy. In both domains, the network bases its clinical decisions on relevant thoracic regions rather than spurious peripheral shortcuts.

5.6. Real-Synthetic Domain Adaptation via DANN

The residual gap motivates an explicit alignment of the real and synthetic distributions with the DANN. As described in Section 4.4, we compare two runs that differ only in the GRL strength λ: the control ($\lambda = 0$, pure augmentation) and the DANN (λ annealed $0 \rightarrow 1$). Table 7 reports the results on the real test set.

Configuration	Normal (F1)	Pneumonia (F1)	Macro F1
Control ($\lambda = 0$, augmentation)	0.66	0.87	0.77
DANN ($\lambda \rightarrow 1$)	**0.74**	**0.89**	**0.81**
Delta	+0.08	+0.02	+0.04

Table 7. Controlled comparison on the real test set: same pipeline with the adversarial alignment disabled ($\lambda = 0$) versus active ($\lambda \rightarrow 1$).

The adversarial alignment gives a small but consistent improvement, driven by the Normal class (F1 $0.66 \rightarrow 0.74$). The control here is below the best augmentation of Table 4 because the adaptation pipeline uses a different training recipe; the fair contrast is therefore $\lambda = 0$ vs $\lambda \rightarrow 1$.

The alignment is only *partial*: the domain-discriminator accuracy falls from ~ 0.81 to ~ 0.72 but never reaches chance (0.5), so the domains are attenuated, not made invariant. This matches the residual separation seen in the PCA/t-SNE/UMAP projections (Section 5.4): rather than erasing the domain difference, the adversarial term applies only a light pressure that acts as a mild regularizer and slightly helps the minority class.

5.7. Cross-Hospital Domain Adaptation

As a final test, we check whether the in-domain benefit of augmentation carries over to a different hospital: we train on Kermany and test on VinDr-PCXR, a pediatric chest X-ray dataset from another country. The two datasets look very different (Figure 6): in VinDr the intensity is reversed, so the lungs are bright and the background is dark, the opposite of Kermany. The class balance is flipped as well: on

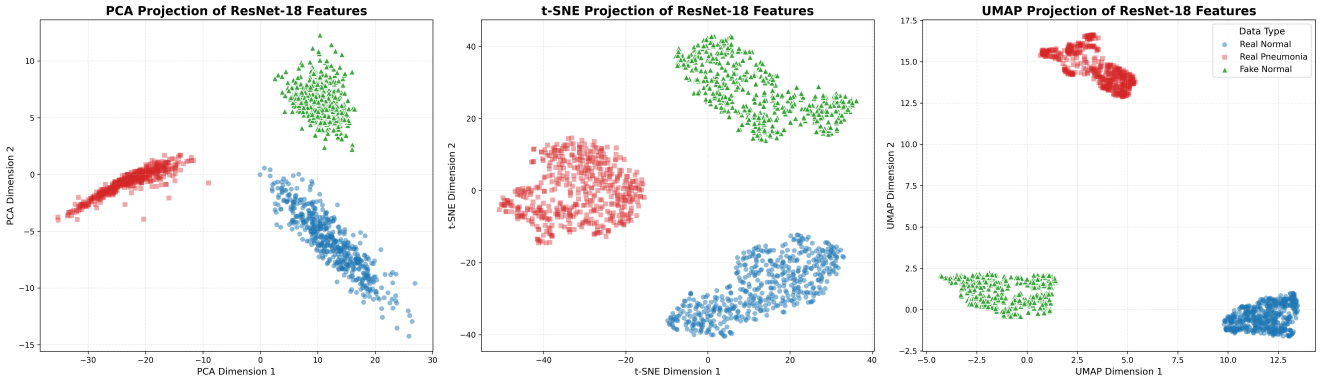


Figure 4. Visualization of the ResNet-18 feature space via PCA (left), t-SNE (center), and UMAP (right) for real and synthetic images.

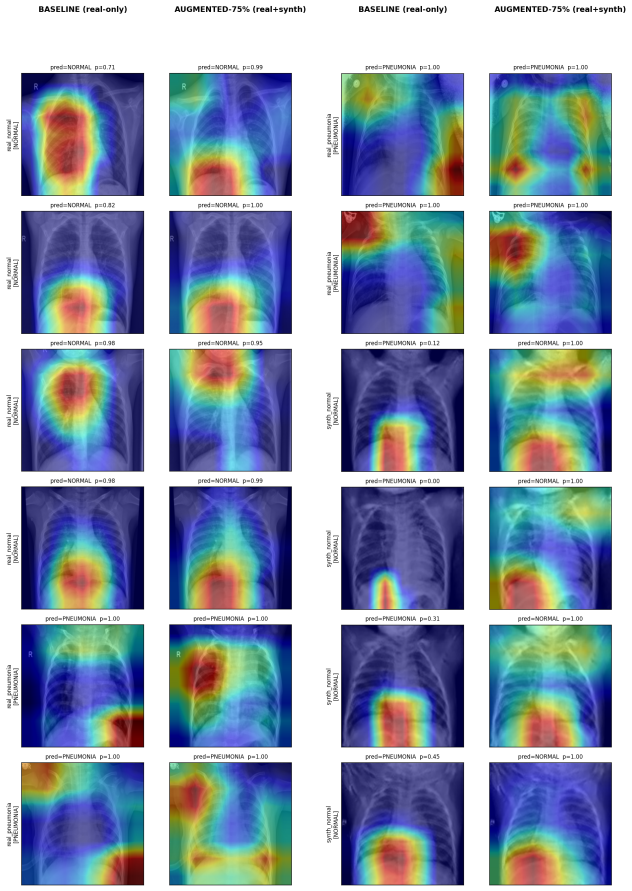


Figure 5. Grad-CAM comparison between the baseline (left) and augmented-75% (right) classifiers on real Normal, real Pneumonia and synthetic Normal samples.

the VinDr test set Normal is the majority (515 images) while Pneumonia is a small minority (92 images). We use unsupervised domain adaptation, so the VinDr labels are used only for the final evaluation, and we compare the original and the augmented (75% gap) source, each with and without DANN. As the main metric we use the Macro F1; in addition, since Pneumonia is the clinically relevant class and, on this target, the rare one, we report its Average Precision (AUPRC), which is the standard metric for detecting a rare positive class under imbalance (the corresponding value for the majority Normal class is close to 1 and uninformative).

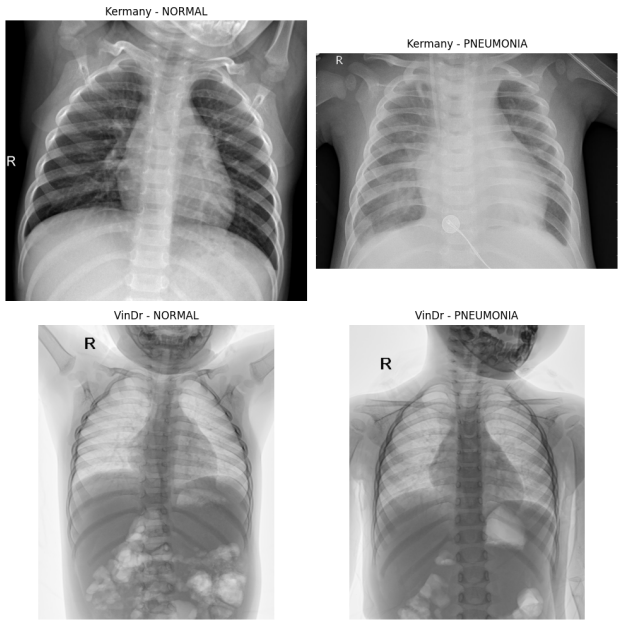


Figure 6. A Normal (left) and a Pneumonia (right) chest X-ray from VinDr-PCXR. The intensity is reversed with respect to Kermany: the lungs appear bright and the background dark.

On the new hospital, nothing we add helps: the plain original source transfers best, and both the synthetic augmentation and the DANN make things worse, step by step (Table 8). Even when the domain discriminator reaches chance accuracy, meaning the two datasets look identical to it, the model still fails to detect pneumonia on VinDr, with an AUPRC close to the ≈ 0.15 we would get by guessing.

Configuration (VinDr test)	Macro F1	AUPRC-Pneu.
Original, source-only	**0.563**	**0.235**
Original, DANN	0.550	0.206
Augmented, source-only	0.509	0.168
Augmented, DANN	0.397	0.129

Table 8. Cross-hospital transfer to VinDr-PCXR. The plain original source transfers best; both generative augmentation and adversarial alignment make it worse.

The reason is intuitive: the GAN learns the specific look of Kermany, including its reversed intensity, so adding its synthetic images makes the classifier even more tuned to the source hospital, exactly the wrong thing when the target looks so different. In short, the gains measured on the same hospital do not carry over to a new one, and simply aligning the feature distributions cannot make up for such different appearance and class balance. Harmonizing the low-level appearance of the two datasets would be a natural next step.

6. Conclusions and Future Developments

Our experiments show that a better test-set F1 does not by itself mean that real and synthetic data are truly aligned. Adding synthetic images improved the minority-class F1, yet the PCA, t-SNE and UMAP projections revealed a clear and persistent separation between real and generated images, while the Grad-CAM maps confirmed that the classifier still relies on the lung regions rather than on generation artifacts. This is why it is worth going beyond a single accuracy number and inspecting how and where the model makes its decisions. Trying to close this gap with adversarial Domain Adaptation (DANN) had only a small and controlled effect on the minority class, and never made the two domains indistinguishable. The cross-hospital test was more telling: on a different institution (VinDr-PCXR), whose images differ from the source even in intensity polarity and class balance, neither augmentation nor DANN helped, and augmentation actually hurt, a case of negative transfer in which the synthetic images make the model more specialised to the source hospital. In short, improvements measured on the same data distribution do not automatically carry over to a new one. Future work includes diffusion models or GANs pre-trained on larger clinical datasets (e.g. CheXpert) for more faithful samples, domain-adaptation methods that reduce the shift while preserving class information, and harmonising the low-level appearance of the images across institutions before transfer.

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